

Profile Photo



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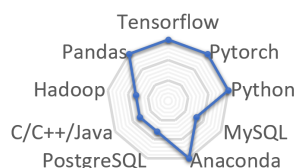
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Personal Links

Homepage
Google Scholar
GitHub
Linkedin

Programming Tools



Personal Skills



Languages

Chinese ★★★★★
English ★★★★★

XinDu

Working Experience

2024-Now	Research Scientist Trustworthy Machine Learning.	Great Bay University, CN
2021-2023	Postdoc Research Associate Trustworthy Autonomous System.	University of Edinburgh, Edinburgh, UK
2020-2021	Postdoc Research Fellow Exceptional Model Mining, Fairness and Causal Inference	Eindhoven University of Technology, Eindhoven, Netherlands
2015-2016	Research Associate 3D City Modeling.	Delft University of Technology, Delft, Netherlands
2015	Algorithm Research Engineer Algorithm design and development for microblog texts labeling.	Weibo, Beijing, China
2010-2012	Software Developer Engineer 4G Map Localization.	Longmap, ShenZhen, China

Research Experience

2017-2020	PhD Research Department of Mathematics and Computer Science	Eindhoven University of Technology, The Netherlands
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- **Fairness in Machine Learning.** Study fairness in terms of unsupervised sensitive subgroups. Propose a fairness measure for network representation model.
- **Causal Inference.** Study the causal effect inference of individual treatments with observational data. Propose a covariate balancing technique to remove confounding bias from imbalanced data.
- **Exceptional Model Mining.** Study the exceptional multi-modal behavior on subgroups. Propose a Bayesian non-parametric model for the inference of exceptional behavior in terms of space, time and texts.
- **Learning Analytics.** Study the heterogeneous learning behavior in MOOCs.

Relevant Research Skills

- **Probabilistic Methods** Formulating problems, building statistical models and making inference from data, using point estimate or Bayesian inference.
- **Latent variable Methods** Modeling generating process and inferring parameters considering the latent variables, using EM, MCMC and variational inference.
- **Hypothesis testing** Formulating hypothesis testing with parametric/non-parametric assumptions. Validating the significance of deviations between two distributions.
- **Meta Learning** Develop learning frameworks that learns meta representation in supervised and un-supervised method.
- **Transfer Learning** Develop machine learning systems that make use of part of the learned dynamic systems that can transfer knowledge from the origin to the target environments.

- **Data mining** Designing efficient data mining tools to discover patterns in particular representations with regard to specific interestingness, e.g. exceptional performance of a model on subgroups.
- **Deep Learning Methods** Building deep neural networks and learning to optimize the performance with data. extracting useful features and applying to downstream tasks like classification and regression.
- **Causal Methods** Employing domain knowledge to build causal graph for underlying generating mechanism. Using causal graph to boost the model on tasks like counterfactual prediction, preventing the misleading of spurious associations.
- **Synthetic Analysis** Design synthetic data with specific generating process to validate particular methods.

Academic Experience

2020-	Program Committee Member	
	ECML-PKDD 2020-2025, IJCAI 2021-2025, AAAI 2021-2026, UAI 2021-2025, AISTATS 2022-2026, IDA 2023, ICLR 2023, KDD 2025, NeurIPS 2025	
2019-	Journal Reviewer	Springer
	International Journal of Artificial Intelligence In Education (IJAIED)	
2019	Proceeding Chair	Würzburg, Germany
	European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)	
2018	Volunteer	's-Hertogenbosch, the Netherlands
	International Symposium on Intelligent Data Analysis (IDA)	
2017	Volunteer	Eindhoven, the Netherlands
	The Annual Machine Learning Conference of The Benelux (Benelearn)	
2019	Academic Visiting	Helsinki University, Finland
	Data Mining Research	

Awards&Copyright&Patent

2020	Student Travel Award	New York, U.S.A
	Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI)	
2014	Software Copyright:2014SR036739	Wuhan University
	Multi-scale map data matching software(GeoMatching)	
2014	Software Copyright:2014SR036893	Wuhan University
	POI visualization based on ubiquitous space and mapping system(POI Viewer)	
2014	Software Copyright:2014SR025346	Wuhan University
	Ocean survey data management and 3D visualization information system.	
2013	Patent	CN 103473420 A
	Automatic positioning method of statistical graph in zonal statistic map	

Education

- 2017 - 2020 **PhD Student in Computer Science** [Eindhoven University of Technology](#)
Supervisor: Prof. dr. Mykola Pechenizkiy (m.pechenizkiy@tue.nl),
Co-Supervisor : Dr. Wouter Duivesteijn (w.duivesteijn@tue.nl),
Thesis: "Uncertainty in Exceptional Model Mining".
- 2012 - 2015 **Master's Degree in GIS** [Wuhan University, China](#)
Supervisor : Prof. dr. Tinghua Ai (tinghuaai@whu.edu.cn),
Thesis: "Information mining of place name based on crowdsourcing data".
Thesis activity carried out during the work period at Weibo, China.
- 2006 - 2010 **Bachelor's Degree in GIS.** [Yunnan University, China](#)
Main subjects: Mathematics, Computer Science, GIS.

Referees

Mykola Pechenizkiy
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Alexander Nikolaev
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Publications

- X. Du, S. Ramamoorthy, W. Duivesteijn, J. Tian, M. Pechenizkiy
Beyond Discriminant Patterns: On the Robustness of Decision Rule Ensembles
The IEEE International Conference on Data Mining (ICDM), 2025.PDF
- X. Du, S. Yang, W. Duivesteijn, M. Pechenizkiy
Conformalized Exceptional Model Mining: Telling Where Your Model Performs (Not) Well
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2025, .PDF
- X. Du, Y. Pei, W. Duivesteijn, M. Pechenizkiy
Exceptional Spatio-Temporal Behavior Mining through Bayesian Non-Parametric Modeling
Data Mining and Knowledge Discovery (ECML-PKDD Journal Track), 2020.PDF
- X. Du, Y. Pei, W. Duivesteijn, M. Pechenizkiy
Fairness in Network Representation by Latent Structural Heterogeneity in Observational Data
AAAI Conference on Artificial Intelligence (AAAI), 2020.PDF
- X. Du, L. Sun, W. Duivesteijn, A. Nikolaev and M. Pechenizkiy
Adversarial Representation Learning for Causal Effect Inference with Observational Data
Data Mining and Knowledge Discovery, 2021.PDF
- X. Du, W. Duivesteijn, M. Klabbers, M. Pechenizkiy
ELBA: Exceptional Learning Behavior Analysis
Proceedings of the Eleventh International Conference on Educational Data Mining (EDM), 2018.PDF

X. Du, B. Legastelois, B. Ganesh, A. Rajan, H. Chockler, V. Belle, S. Anderson, S. Ramamoorthy

Vision Checklist: Testable Error Analysis of Image Models to Help System Designers Interrogate Model Capabilities *Arxiv, 2022.PDF*

Y. Pei, X. Du, J. Zhang, G. Fletcher, M. Pechenizkiy

struc2gauss: Structure Preserving Network Embedding via Gaussian Embedding *Data Mining and Knowledge Discovery, 2020.PDF*

Y. Pei, X. Du, J. Zhang, G. Fletcher, M. Pechenizkiy

Dynamic Network Representation Learning via Gaussian Embedding *Graph Representation Learning Workshop (NeurIPS), 2019.*

Anthony L. Corso, Sydney M. Katz, Craig Innes, Xin Du, Subramanian Ramamoorthy, Mykel J. Kochenderfer

Risk-Driven Design of Perception Systems *The 36th Conference on Neural Information Processing Systems, 2022.PDF*

Software

ABCEI

Developer

A causal inference software to estimate CATE

[Causal Inference](#)

Annotate Optimize**Developer**

A software to optimize map annotation

[Map Annotation](#)